

The Weyl energy of cancellation, reduced

Exact Fourier decomposition of the three open objects, Parseval on the cosine grid, and the scale that actually closes

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August 28, 2026

Abstract

Four majorant routes died at the same place (r381 L^∞ , r385 Koksma, r387 Gershgorin, r388 osc-Geronimus). The three open cancellation objects are the signed μ -CD off-diagonal (assist 0.0399 against Gershgorin 13.32), terminal C_ε ($|\text{FO}| \leq \frac{1}{32}$ on the last twelve), and Z_{loc} (median cancel 0.055, not uniform). This note replaces majorization by an exact Fourier identity. What is proved: on the cosine grid the Chebyshev- T Christoffel–Darboux kernel is a finite cosine sum, and

$$\Phi_k := \sum_{i,j} v_i v_j K_k^T(x_i, x_j) = \frac{1}{\pi} |S_0|^2 + \frac{2}{\pi} \sum_{m=1}^{k-1} C_m^2,$$

with $S_q = \sum_j v_j e^{iq\theta_j}$ and $C_m = \text{Re } S_m$; the off-diagonal is Φ_k minus an explicit v^2 -Weyl diagonal; $Q_k^T = (C_0 + C_{2k})_\nu / (C_0 + C_{2k})_\mu$; π_k^2 is a finite combination of $C_0, \dots, C_{2k}$ via the Chebyshev coefficients of π_k ; and the finite-grid Parseval $\sum_{q=0}^{2S-1} |S_q|^2 = 2S \sum_j v_j^2$ holds with ratio 1 on the two-period mesh and on FRAME-A $w = 9$. What is *not* proved: a bound on $\sum c_q |S_q|^2$ that closes all three objects. MAIN and χ_3 sit at quadratic-mean ($\text{mean}_{m \geq 1} C_m^2 / (\frac{1}{2} \sum v^2) \in [0.66, 1.02]$ on core-42); the two-period is a spectral comb ($\rho_{\text{AP}} = 1$ if and only if the energy concentrates on a progression); scramble holds quadratic-mean and dies on the μ -OP remainder. The Chebyshev piece of C_ε closes at quadratic-mean ($\text{FO}^T \leq 0.0165 < \frac{1}{32}$ on the eleven small- N windows). Assist is not the Chebyshev all-ones energy ($1^T E^T 1 / n_\nu = 0.114$ against $\lambda = 0.99983$). Z_{loc} is not the Chebyshev xT_k^2 proxy (cancel 0.47). The hydra-legal remainder is the Weyl energy of the μ -OP Chebyshev-coefficient Gram: of ϕ^* (assist), of consecutive α_k ($d\Delta / C_\varepsilon$), of the edge-masked drive (Z_{loc}). No RH claim.

Status. Lemmas 1–4 are proved (finite identities). Lemma 2 and Lemma 3 are proved (finite identities). Proposition 1 is a finite machine census. Proposition 2 records the scale adjudication. Proposition 3 names the reduction. No RH claim. No L^* claim. No C_ε claim.

1 The object

Write $\theta = \arccos x$ on the cosine grid of [4], $S_q = \sum_j v_j e^{iq\theta_j}$ for the ν -weights, and $C_m = \sum_j v_j \cos(m\theta_j) = \text{Re } S_m$. The Chebyshev- T CD kernel of k orthonormal functions is

$$K_k^T(\cos \theta, \cos \phi) = \frac{1}{\pi} + \frac{2}{\pi} \sum_{m=1}^{k-1} \cos(m\theta) \cos(m\phi) = \frac{D_{k-1}(\theta - \phi) + D_{k-1}(\theta + \phi)}{2\pi}, \quad (1)$$

the r387 identity. The IKS Gram is $(E_k)_{ij} = \sqrt{v_i v_j} K_k^\mu(y_i, y_j)$ with K_k^μ the μ -CD kernel; r285 writes $\lambda_{\text{max}}(E_k) = \text{maxdiag}_k \cdot (1 + \text{assist}_k)$. Round 385: $\text{FO}_k[-\nu] = \gamma_k(Q_k - Q_{k-1})$ with $Q_k = Q_k^T + \Delta_k$. Z_{loc} is the edge piece of the r383 canonical split.

The candidate of this round: compute the three cancellations as explicit linear combinations of C_m (or $|S_q|^2$), then read off the scale of a closing bound from the measured spectrum.

2 What is exact

Lemma 1 (Chebyshev energy). *For any atomic weights v on $(-1, 1)$,*

$$\Phi_k = \sum_{i,j} v_i v_j K_k^T(x_i, x_j) = \frac{1}{\pi} C_0^2 + \frac{2}{\pi} \sum_{m=1}^{k-1} C_m^2.$$

The off-diagonal is $\Phi_k - \sum_i v_i^2 K_k^T(x_i, x_i)$, with

$$K_k^T(x, x) = \frac{k}{\pi} + \frac{1}{\pi} \sum_{m=1}^{k-1} \cos(2m\theta), \quad \sum_i v_i^2 K_k^T(x_i, x_i) = \frac{k}{\pi} \sum_i v_i^2 + \frac{1}{\pi} \sum_{m=1}^{k-1} W_{2m},$$

$W_q = \sum_i v_i^2 \cos(q\theta_i)$. *Exact over $\mathbb{Q}$ on a five-atom toy ($\Phi^* = 99$, off = 26, diag = 73, the $1/\pi$ scale stripped). Floating-point six-atom residual 0 (energy) and $3.4 \cdot 10^{-16}$ (off). FRAME-A $w = 9$, $k = 8$: residual $2.6 \cdot 10^{-15}$; mpmath dps 30 residual $4.3 \cdot 10^{-31}$.*

The μ -CD remainder is carried explicitly: $K_k^\mu = K_k^T + R_k$, so the $v \otimes v$ bilinear splits as $\Phi_k^\mu = \Phi_k + \sum_{i,j} v_i v_j R_k$. At $k = 22$ on $w = 9$ this remainder is -0.0028 against $\Phi = 0.026$ (small). The $\sqrt{v}$ -dressed Gram E is a *different* quadratic form: $1^T E 1 = 16.05$ against $1^T E^T 1 = 6.44$ at the same k (remainder of order one).

Lemma 2 (Q^T as a Weyl ratio).

$$Q_k^T = \frac{C_0(\nu) + C_{2k}(\nu)}{C_0(\mu) + C_{2k}(\mu)}.$$

This is [2]'s Chebyshev sampling, rewritten in C_m . Checked on a six-atom toy at $k = 0, 1, 2, 4$ (residual 0) and on FRAME-A $w = 9$ at $k = 0, 10, 50, 182$.

Lemma 3 (π_k^2 Weyl). *If $\pi_k = \sum_{m=0}^k \alpha_m T_m$ in the Chebyshev basis, then*

$$Q_k = \sum_j v_j \pi_k(x_j)^2 = \sum_{p,q=0}^k \alpha_p \alpha_q \cdot \frac{1}{2} (C_{p+q} + C_{|p-q|}).$$

Five-atom toy, degree 3, residual $5 \cdot 10^{-16}$. Consequently $\text{FO}_k = \gamma_k(Q_k - Q_{k-1})$ is a linear combination of $C_0, \dots, C_{2k}$, plus the r388 split into $dQ^T + d\Delta$.

Lemma 4 (finite-grid Parseval). *On the cosine index grid $\theta_j = \pi j/S$, $j = 1, \dots, S$, for any (possibly sparse) real weights v ,*

$$\sum_{q=0}^{2S-1} |S_q|^2 = 2S \sum_j v_j^2.$$

Ratio = 1 at machine precision on the two-period mesh $S = 21$ and on FRAME-A $w = 9$ (nu-only and the kept union of 367 nodes).

This is the exact Parseval balance asked of the round. It does *not* by itself imply quadratic-mean per mode: the two-period saturates Parseval *and* concentrates.

3 The spectrum

Write $qm := \frac{1}{2} \sum v^2$ for the quadratic-mean reference ($E[C_m^2]$ if a mode were a random phase on the circle) and $HHI(C) = \sum_{m=1}^M p_m^2$ for the energy share $p_m \propto C_m^2$.

Proposition 1 (census). *On the sealed constructors:*

1. *FRAME-A* $w = 9$, $M = 400$: $\text{mean}/qm = 0.922$, $\text{max}/qm = 14.5$ at $m = 208$, $|C|/\text{mass}$ last-twelve in $[0.0005, 0.179]$ (trivial Weyl, not PNT), $HHI_{1..80} = 0.053$ (random ~ 0.013).
2. *Core-42*, $M = 200$: $\text{mean}/qm \in [0.661, 1.018]$, $|C|/\text{mass}$ max in $[0.229, 0.401] < \frac{1}{2}$, $HHI \in [0.013, 0.097] < 0.12$.
3. *EXT-heavy seven*: $|C|/\text{mass}$ max in $[0.237, 0.348] < 0.40$; $HHI \in [0.16, 0.35]$ is flank-mass, not a two-period comb.
4. χ_3 -42: $\text{mean}/qm \in [0.658, 0.909]$, same class as MAIN.
5. Two-period $S = 81$, $c = \frac{2}{3}$: $|C_S|/\text{mass} = 1$, $HHI_{1..80} = 0.727$, comb at $q = 0, 1, S - 1, S, S + 1$.
6. Scramble seed 1: $\text{mean}/qm = 0.947$ (quadratic-mean holds), $HHI = 0.007$ (more uniform than MAIN). Top $|\Delta C_m^2|$ modes 380, 354, 30, 177, 170, 23.
7. Von Mangoldt versus equal weight on $w = 9$, $M = 200$: weighted $\text{mean}/qm = 0.718 > \text{equal } 0.403$. The weights do not buy extra cosine cancellation.

The cancellation of the program, at the Chebyshev layer, sits at quadratic-mean: no structured valley below qm , no resonance at a fold frequency on MAIN. The two-period is the concentrating adversary, and $\rho_{AP} = 1$ is exactly that comb. Scramble does not die on ν -Weyl concentration.

4 Scale adjudication

Proposition 2 (scale). (a) Assist. Need assist $< (1 - \text{maxdiag})/\text{maxdiag} = 0.0401$ at the $w = 9$ wall (true 0.0399). The Chebyshev all-ones Rayleigh is $1^T E^T 1/n_\nu = 0.114 \ll \lambda = 0.99983$; the $v \otimes v$ Chebyshev off-diagonal at the wall is negative (-0.007). A quadratic-mean majorant of $1^T E^T 1$ overshoots the actual Chebyshev energy by 1.52. Quadratic-mean of the Chebyshev bilinear does not close assist: it is the wrong quadratic form (all-ones, not the top mode of E).

(b) C_ε . Need $|\text{FO}| \leq \frac{1}{32}$ on the last twelve. $\text{FO}^T = 0.00496 < \frac{1}{32}$ on $w = 9$; eleven small- N windows give $\max|\text{FO}^T| = 0.01650$ and $\max|\text{FO}| = 0.01182$, both below the bar. The Chebyshev piece closes at quadratic-mean ($|C_{2k}|/\text{mass}$ last-twelve mean 0.089 against rms-prediction 0.105). The live remainder is $d\Delta$ (r388).

(c) Z_{loc} . The Chebyshev proxy $\sum v \cos \theta (1 + \cos 2k\theta)/2$ has signed/abs cancel 0.475, against r386's median 0.055. The edge restriction $|x| > 0.85$ is worse (0.86). Z_{loc} is the edge-masked drive of the actual μ -OP, not this proxy; quadratic-mean of the proxy does not close uniform cancel.

PNT never enters: $|C|/\text{mass} \leq 0.40$ is already the trivial Weyl size, and the Chebyshev FO^T bound does not demand anything smaller.

5 Reduction

Proposition 3 (reduced). *The energy identity and the finite-grid Parseval are theorems. MAIN/ χ are at quadratic-mean; $\rho_{\text{AP}} < 1/5$ is the time-domain form of “not a comb”. The deduction “therefore assist, C_ε and Z_{loc} close at quadratic-mean” is false for assist and Z_{loc} , and true for the Chebyshev piece of C_ε . The hydra-legal remainder is the Weyl energy of the μ -OP Chebyshev-coefficient Gram: of the top eigenvector ϕ^* of E (assist), of consecutive Chebyshev coefficients α_k of π_k ($d\Delta$, the live C_ε), and of the edge-masked drive (Z_{loc}).*

6 Kills

Wrong c_q (1 instead of 2) fails at relative error 0.108. Dropping the diagonal fails at $|\Phi - \text{off}|/|\text{off}| = 2.78$. The two-period is visible as a q -comb ($|C_S|/\text{mass} = 1$, $\text{HHI} = 0.727$); this is ρ_{AP} as a spectral statement. Scramble seed 1 holds quadratic-mean and is *more* uniform than MAIN; the divergence modes of $|\Delta C_m^2|$ are 380, 354, 30, 177, 170, 23. It dies on the μ -OP remainder ($d\Delta = 2.16 \cdot 10^6$ of [5]), not on ν -Weyl concentration.

References

- [1] S. Hamann, *The G_ε lemma, reduced*, rh/problem/g_eps_lemma.tex, 2026.
- [2] S. Hamann, *Christoffel quietness, reduced*, rh/problem/christoffel_quiet.tex, 2026.
- [3] S. Hamann, *Living-ladder (Z') and cofinal R_0* , rh/problem/compose_premises2.tex, 2026.
- [4] S. Hamann, *Coherence assist, reduced*, rh/problem/coherence_assist.tex, 2026.
- [5] S. Hamann, *The Δ -deformation, reduced*, rh/problem/delta_deformation.tex, 2026.